COMBINATORICS
EXAM 1

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1. Answer the following using simple counting reasoning

(a) How many different 7-digit numbers (non-negative integers) are there? (Leading zeros are not allowed).
(b) How many even 7-digit numbers are there?
(c) How many 7-digit numbers are there with exactly one 7?
(d) How many 7-digit numbers read the same when you read themm forward or backward, i.e. are palindromic, e.g. 6548456 is one such integer.
(e) How many 7-digit numbers are there that are divisible by 4?
(f) How many times is the digit 7 written when listing the numbers between 1 and 1,000,000?

Solution. (a) If we consider the number simply as a string of characters, and we allowed "0" at the beginning, then we would have 10^7 solutions, but for our leading digit we omit the availability of 0. So our count becomes 9×10^6.

(b) To get the number of 7-digit even numbers with the same rules, we simply take the result from (a) and divide by 2. So we get our count to be $(9 \times 10^6)/2 = 4.5 \times 10^6$.

(c) Let's approach the problem in cases. Case 1 (the 7 is the first digit): the other 6 digits are anything but 7 which is 9^6 options. Case 2 (the 7 is in one of the other 6 positions): we start with the first digit which is 8 choices, because it can't be 0 or 7. Next we fill the remaining 5 non-7 digits in 9^5 ways. Lastly, we have to choose which of the 6 digits is our 7, and there are 6 ways to do that. So the count of 7 in one of the other 6 positions comes out as $8 \cdot 9^5 \cdot 6$.

(d) Considering the palindromes, we need only consider half the word because the other half mirrors it, excepting the middle character. Thus, our count for ways to have a palindrome becomes 9×10^3.

(e) At first glance I went about simply dividing (a) by 4, but that over simplifies the task at hand and omits some subtleties. We begin by noticing that if a number is divisible by 4 then the last two digits form a number that is divisible by 4.

So let's break the number down into $d_1d_2d_3d_4d_5d_6d_7$. To build our number we chose the first digit d_1 between 1 and 9 to remain a 7-digit number. The middle digits can be

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anything from 0 to 9 which gives us 10^4 choices. Lastly, the last two digits must be in the set $\{00, 04, 08, \ldots, 96\}$ which has 25 elements. So the number of choices we have to build our number is $9 \cdot 25 \cdot 10^4 = 2250000 = 2.25 \times 10^6$

(f) Notice that $1,000,000$ can not contain a 7 so we only concern ourselves with numbers of length 6. In each of the 6 positions a 7 occurs in 10^5 numbers, 6×10^5. And, because 0 and $1,000,000$ don't contain the digit 7 we're good with our count being 6×10^5. □

2. How many 7-letter words can be constructed by using the 26 letters of the alphabet if each word contains 3, or 4 vowels? It is understood that there is no restriction on the number of times a letter can be used in a word.

Solution. Let $\{a, e, i, o, u, y\}$ be our set of vowels. The number of 7-letter words with 3 vowels is

$$\binom{7}{3} \cdot 6^3 \cdot 20^4.$$

The number of 7-letter words with 4 vowels is

$$\binom{7}{4} \cdot 6^4 \cdot 20^3.$$

So our count follows as:

$$\# = \binom{7}{3} \cdot 6^3 \cdot 20^4 + \binom{7}{4} \cdot 6^4 \cdot 20^3 = 1572480000$$

□

3. Let n and k be positive integers. Give a combinatorial proof that

$$\sum_{k=1}^{n} k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$$

Proof. We can reduce this problem to a question about a committee with men and women. And we want to choose n committee members from a pool of n men and n women, and we want a woman chairperson. The right hand side is clearly a count of just this. For the left hand side consider a committee we choose k women from n choices, $\binom{n}{k}$. Then, we count the $n-k$ men we chose from the n men $\binom{n}{n-k} = \binom{n}{k}$. All we are left to do is select woman who is supposed to chair the committee, and there are n ways to accomplish this task. And thus we clearly have that:

$$\sum_{k=1}^{n} k \binom{n}{k}^2 = n \binom{2n-1}{n-1}$$

□

4. In how many ways can we select a set of m marbles from a collection of $2m$ marbles, where m of them are identical and the remaining m are all distinct?

Solution. Let the collection of marbles look like $\{\underbrace{l, l, \ldots, l}_{m-\text{terms}}, m_1, m_2, \ldots, m_m\}$. Choose r distinct marbles where $0 \leq r \leq$ in $\binom{m}{r}$ ways. Then we must choose the identical marbles in $m - r$ ways, for which there is only 1 way to do this since they are all identical. Thus, the number of ways to do our selection generally is:

$$\sum_{r=0}^{m} \binom{m}{r} = 2^m.$$

□

5. Show that if $n+1$ distinct integers are chosen from the set $\{1, 2, \ldots, 3n\}$, then there are always two which differ by at most 2.

Proof. For the sake of curiosity, let's look at the partition of our set in n distinct blocks of three consecutive integers. From these blocks we choose $n+1$ integers, so by the pigeonhole principle one of the blocks must contain two elements. But, any block that contains two elements contains two numbers that differ by no more than two, and we are done. □

6. Use the pigeonhole principle to show that:

(a) Any set of 7 distinct integers includes two integers a and b such that either $a+b$ or $a-b$ is a multiple of 10.

(b) Any group of 20 students either has a subgroup of 4 mutual acquaintances, or a subgroup of 4 mutual strangers.

Solution. (a) We begin by using the mechanics of the integers modulo 10 to sketch out the pigeonholes that we are working with. We begin by noticing that we end up with 10 "residues" of the integers modulo 10. In other words, any integer can be written as

$$n \equiv r \mod 10, \quad \text{with} \quad r \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}.$$

Futhermore we are in one of two situations if

$$a \equiv b \mod 10 \Rightarrow a - b \equiv 0 \mod 10.$$

or where

$$a + b \equiv 10 \equiv 0 \mod 10.$$

We then see that we have 6 possible different sets of integers into which our pick of seven, from the problem statement, must land:

1. $S_0 = \{0\}$, if two numbers have remainder 0, their difference is a multiple of 10
2. $S_1 = \{1, 9\}$, sum is 10, or difference is 0, similarly
3. $S_2 = \{2, 8\}$, where the sum is 10, or difference is 0,
4. $S_3 = \{3, 7\}$, sum is 10, or difference 0,

5. $S_4 = \{4, 6\}$, sum is 10, or difference is 0, and
6. $S_5 = \{5\}$, which if two numbers have remainder 5, their sum $5+5 = 10$ or their difference $5 - 5 = 0$.

So now we are choosing seven distinct integers that must land in our six distinct sets. Let these two integers be a and b. If a and b share the same remainder when they land in our set S_i, we have $a - b \equiv 0 \mod 10$. On the other had if they don't share the same remainder when they land in our set then $a + b \equiv 10 \mod 10$. So by definition of the integers modulo 10 we have that either $10|(a - b)$ or $10|(a + b)$.

(b) For 18 we rely upon the Ramsey number for $R(4, 4) = 18$. For the sake of argument we reduce the problem to a graph coloring problem: we want a K_{20} with a two coloring (red/blue for example) such that there is a monochromatic subgraph K_4 (in either red or blue). Well we know that for K_{18} such a property necessarily exists and adding 2 more vertices (or students) to the equation will continue to have that monochromatic K_4. □

7. What is the number of integral solutions of the equation

$$x_1 + x_2 + x_3 + x_4 = 30,$$

such that $x_1 \geq 1$, $x_2 \geq 2$, $x_3 \geq 3$, and $x_4 \geq 4$?

Solution. We begin by shifting to 0 by setting $y_1 \geq x_1 - 1$, $y_2 \geq x_2 - 2$, $y_3 \geq x_3 - 3$, and $y_4 \geq x_4 - 4$, with $y_1 \geq 0$, $y_2 \geq 0$, $y_3 \geq 0$, and $y_4 \geq 0$. thus our equation becomes

$$\begin{aligned} (y_1 + 1) + (y_2 + 2) + (y_3 + 3) + (y_4 + 4) &= 30 \\ y_1 + y_2 + y_3 + y_4 + 10 &= 30 \\ y_1 + y_2 + y_3 + y_4 &= 20. \end{aligned}$$

So the number of possible solutions follows as

$$\binom{20 + 4 - 1}{4 - 1} = \binom{23}{3} = 1771$$

□

8. Find the number of integer solutions to the equation

$$x_1 + x_2 + x_3 + x_4 = 20,$$

for which $1 \leq x_1 \leq 6$, $0 \leq x_2 \leq 7$, $4 \leq x_3 \leq 8$, and $2 \leq x_4 \leq 6$?

Solution. For this problem we approach it much the same as the last problem regarding the lower bounds which we shift to zero with the relations: $0 \leq y_1 \leq 5$, $0 \leq y_2 \leq 7$, $0 \leq y_3 \leq 4$, and $0 \leq y_4 \leq 4$. And the function becomes:

$$\begin{aligned}(y_1+1)+y_2+(y_3+4)+(y_4+2) &= 20\\ y_1+y_2+y_3+y_4+7 &= 20\\ y_1+y_2+y_3+y_4 &= 13\end{aligned}$$

So our initial count begins as:

$$\binom{13+4-1}{4-1} = \binom{16}{3} = 560.$$

But we have potentially double counted, triple counted, and quadruply counted solutions that lie outside our bounds.

So, remembering that we have $0 \leq y_1 \leq 5$, $0 \leq y_2 \leq 7$, $0 \leq y_3 \leq 4$, and $0 \leq y_4 \leq 4$ we consider the exclusion sets

$$A_1 = \{y_1 \geq 6\}, \quad A_2 = \{y_2 \geq 8\}, \quad A_3\{y_3 \geq 5\}, \quad \text{and} \quad A_4 = \{y_4 \geq 5\}.$$

To remove our double countings, we count our excluson sets from above

- A_1: $(y_1+6)+y_2+y_3+y_4 = 13 \Rightarrow \binom{7+4-1}{4-1} = \binom{10}{3} = 120$,
- A_2: $y_1+(y_2+8)+y_3+y_4 = 13 \Rightarrow \binom{5+4-1}{4-1} = \binom{8}{3} = 56$,
- A_3: $y_1+y_2+(y_3+5)+y_4 = 13 \Rightarrow \binom{8+4-1}{4-1} = \binom{11}{3} = 165$, and
- A_4: same as A_3, $\binom{11}{3} = 165$.

So we will need to subtract $120+56+165+165 = 506$ from our original answer.

We will then need to re-add our triple countings in to our main count. So again we resort to cases:

- $A_1 \cap A_2$: $(y_1+6)+(y_2+8)+y_3+y_4 = 13 \Rightarrow \binom{-1+4-1}{4-1} = \binom{2}{3} = 0$,
- $A_1 \cap A_3$: $(y_1+6)+y_2+(y_3+5)+y_4 = 13 \Rightarrow \binom{2+4-1}{4-1} = \binom{5}{3} = 10$,
- $A_1 \cap A_4$: $(y_1+6)+y_2+y_3+(y_4+5) = 13 \Rightarrow \binom{2+4-1}{4-1} = \binom{5}{3} = 10$,
- $A_2 \cap A_3$: $y_1+(y_2+8)+(y_3+5)+y_4 = 13 \Rightarrow \binom{0+4-1}{4-1} = \binom{3}{3} = 1$,
- $A_2 \cap A_4$: $y_1+(y_2+8)+y_3+(y_4+5) = 13 \Rightarrow \binom{0+4-1}{4-1} = \binom{3}{3} = 1$, and
- $A_3 \cap A_4$: $y_1+y_2+(y_3+5)+(y_4+5) = 13 \Rightarrow \binom{3+4-1}{4-1} = \binom{6}{3} = 20$.

And, we notice for any quadruple countings or quintuple countings of the solutions of the original equation we would have an equation where the right hand side would look like $13-(6+5+5) = -3$, $13-(8+5+5) = -5$, $13-(8+6+5) = -6$, or $13-(6+8+5+5) = -11$, none of which could happen.

Thus, our final count is

$$560 - 506 + (10+10+1+1+20) = 96.$$

$\square$

9. Solve the recurrence relation

$$a_n = 5a_{n-1} + 2a_{n-2} - 24a_{n-3}, \qquad n \geq 3$$

given that $a_0 = 1$, $a_1 = 2$, and $a_2 = 6$.

Solution. We begin by writing the recurrence relation as:

$$a_n - 5a_{n-1} - 2a_{n-2} + 24a_{n-3} = 0.$$

For this we write the characteristic polynomial

$$\begin{aligned} \alpha^n - 5\alpha^{n-1} - 2\alpha^{n-2} + 24\alpha^{n-3} &= 0 \\ \alpha^{n-3}(\alpha^3 - 5\alpha^2 - 2\alpha + 24) &= 0 \\ \alpha^{n-3}(\alpha+2)(\alpha^2 - 7\alpha + 12) &= 0 \\ \alpha^{n-3}(\alpha+2)(\alpha-3)(\alpha-4) &= 0. \end{aligned}$$

Thus, we have that $\alpha = -2, 3, 4$, and we can write that $a_n = c_1(-2)^n + c_2 3^n + c_3 4^n$. So we have

$$\begin{aligned} a_0 &= c_1 + c_2 + c_3 = 1 \\ a_1 &= -2c_1 + 3c_2 + 4c^3 = 2 \\ a_2 &= 4c_1 + 9c_2 + 16 = 6 \end{aligned}$$

which sets up as

$$\begin{aligned} \begin{bmatrix} 1 & 1 & 1 \\ -2 & 3 & 4 \\ 4 & 9 & 16 \end{bmatrix} \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} &= \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} \\ \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} &= \begin{bmatrix} 1 & 1 & 1 \\ -2 & 3 & 4 \\ 4 & 9 & 16 \end{bmatrix}^{-1} \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} \\ \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} &= \begin{bmatrix} \frac{2}{5} & -\frac{7}{30} & \frac{1}{30} \\ \frac{8}{5} & \frac{2}{5} & -\frac{1}{5} \\ -1 & -\frac{1}{6} & \frac{1}{6} \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 6 \end{bmatrix} \\ \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} &= \begin{bmatrix} \frac{1}{6} \\ \frac{6}{5} \\ -\frac{1}{3} \end{bmatrix} \end{aligned}$$

$\square$

10. Solve the recurrence relation $a_k = a_{k-1} + k + 6$, with $k \geq 1$ given that $a_0 = 5$.

Solution. We begin by noticing that

$$\begin{aligned} a_1 &= a_0 + 1 + 6 \\ a_2 &= a_1 + 2 + 6 \\ a_3 &= a_2 + 3 + 6 \\ &\vdots \\ a_k &= a_{k-1} + \sum_{k=1}^{k}(j+6). \end{aligned}$$

If we split the sum apart we see that

$$\begin{aligned} \sum_{k=1}^{k}(j+6) &= \sum_{k=1}^{k} j + \sum_{k=1}^{k} 6 \\ &= \frac{k(k+1)}{2} + 6k. \end{aligned}$$

So we further see that:

$$\begin{aligned} a_k &= a_{k-1} + \frac{k(k-1)}{2} + 6k \\ &= \frac{k^2 + 13k + 10}{2} \end{aligned}$$

□

11. Use generating functions to count the nunber of ways to collect \$15 from 20 different students, if each of the first 19 students can give either \$1 or nothing, and the twentieth student can give nothing, \$1, or \$5.

Solution. We begin byu building our solution using ordinary generating functions. Each of the first 19 students contribute either 0 or 1 dollars, so we need: $(1+x)^{19}$. Secondly the 20^{th} student can give 0, 1, or 5 dollars, so we need: $1 + x + x^5$. Thus, when we multiply the parts together we get our generating function

$$\begin{aligned} g(x) &= (1+x)^{19}(1+x+x^5) \\ &= (1+x)^{19} + x(1+x)^{19} + x^5(1+x)^{19}. \end{aligned}$$

So we are going to use a new notation here. Let $[x^n]p(x)$ be the coefficient for x^n in the polynomial $p(x)$. So for example $[x^2](1 + 10x - 3x^2 + 12x^3 + 17x^4) = -3$.

We are primarily concerned with $[x^{15}]g(x)$, as that is what corresponds to the number of ways to collect \$15. So we look at

$$[x^{15}]g(x) = [x^{15}](1+x)^{19} + [x^{15}]x(1+x)^{19} + [x^{15}]x^5(1+x)^{19}$$

But, notice $[x^{15}]x(1+x)^{19} = [x^{14}](1+x)^{19}$ and $[x^{15}]x^5(1+x)^{19} = [x^{10}](1+x)^{19}$. So then we're faced with

$$[x^{15}]g(x) = [x^{15}](1+x)^{19} + [x^{14}](1+x)^{19} + [x^{10}](1+x)^{19}.$$

And by the binomial expansion we have that

$$(1+x)^{19} = \sum_{k=0}^{19} \binom{19}{k} x^k$$

So we have that,

$$[x^{15}]g(x) = \binom{19}{15} + \binom{19}{14} + \binom{19}{10} = 107,882.$$

$\square$

12. Consider an n-sided polygon (with n vertices and n sides).

(a) Assuming no three diagonals intersect at a single point, count the points of intersection formed by the diagonals of this n-sided polygon.

(b) How many line segments are formed by the intersection of points in part (a)?

(c) One of the simplest and most elegant formulae in mathematics is Euler's polyhedron formula. A version for planar graphs states that if a planar graph with v vertices and e edges is drawn in the plane without any edge crossings and has r regions, then

$$r = e - v + 2$$

Use this formula together with your answers in parts (a) and (b) to count the number of regions formed by the diagonals of the n-sided polygon.

Solution. (a) The total number of intersection points is $\binom{n}{4}$. (b) The number of line segments is $\square$

13. Show that any set of 3 distinct integers includes two integers x and y such that

$$F(x,y) = x^3y - xy^3$$

is a multiple of 10.

Proof. First factor $F(x,y)$ as

$$F(x,y) = x^3y - xy^3 = xy(x^2 - y^2) = xy(x+y)(x-y)$$

$F(x,y)$ must always be even because if x, y are both even then both of the terms are even. If x, y are even and odd then xy is even so $F(x,y)$ is even. If both x, y are odd, then x^2 is odd and y^2 is odd, but then $(x^2 - y^2)$ is even. So for every pair $2|F(x,y)$. Thus, to get a multiple of 10, it only remains to show that $5|F(x,y)$. Let's work with the integers modulo

5, and consider the residues $\pmod 5$ grouped by $\pm$ as

$$\{0\}, \quad \{1,4\}, \quad \{2,3\}.$$

When you take an arbitrary three integers $a, b, c \pmod 5$, you get that two of the integers, call them x and y end up such that either

- both reduce to $0 \pmod 5$, and then $5|x$
- they are both in $\{1,4\}$ or both in $\{2,3\}$ which means that $x \equiv y \pmod 5$ $x \equiv -y \pmod 5$. In either case $x-y \equiv 0 \pmod 5$ or $x+y \equiv 0 \pmod 5$, and hence $5|F(x,y)$.

So we have shown that $5|F(x,y)$, and we are done. $\square$

14. Use combinatorial reasoning to show that:

$$\binom{n}{r-1} = \sum_{i=0}^{k} (-1)^i \binom{k}{i} \binom{n+k-i}{r+k-i}$$

Proof. We go about this by using the principle of inclusion-exclusion. Consider a set A containing $n+k$ elements, where k elements are labled $x_1, x_2, \ldots, x_k$, and remaining n elements are labled $y_1, y_2, \ldots, y_n$. We want to choose a subset of size $r+k$ from S such that all k elements $\{x_1, \ldots, x_k\}$ are included. Selecting k specific elements leaves us to choose $(r+k)-k = r$ elements from the remaining n. However, to count all the elements we need to consider the complement and counting subsets of size $r+k$ that contain at least one specific subset of “forbidden” elements.

Next we apply inclusion-exclusion. Let B bet the collection of elements of all subsets of A with size $r+k$. The total number of subsets is $\binom{n+k}{r+k}$. For a fixed set of i excluded elements we must consider $r+k$ elements from the remaining $(r+k)-i$ elements, and the number of ways to do this is $\binom{n+k-i}{r+k-i}$, and there are $\binom{k}{i}$ ways to choose i elements to exclude. So the sum becomes

$$\sum_{i=0}^{k} (-1)^i \binom{k}{i} \binom{n+k-i}{r+k-i}.$$

So how do we get the left hand side of the equation? If we must include all k elements $\{x_1, \ldots, x_k\}$ in our subset of size $r+k$, we are actively choosing r elements from the remaining n elements $\binom{n}{r}$. By the symmetry of Pascal’s identity, the alternating sum of binomial coefficients of this form collapses. Specifically using the identity we see that $\binom{n}{r} = \binom{n-1}{r} + \binom{n-1}{r-1}$, repeatedly k times which equates to $\binom{n}{r-k+k-1} = \binom{n}{r-1}$, the left hand side, and we are done. $\square$

15. Find and verify a general formula, involving the Sterling Numbers of the second kind, for $1^n + 2^n + 3^n + \cdots + m^n$.

Solution. Let U be the set of all partitions of $[p]$ into k distiguishable boxes $B_1, \ldots, B_k$. We define all k properties $P_1, \ldots, P_k$ where P_i is the property that the i^{th} box B_i is empty. Let

A_i denote the subset of U consisting of all of those partitions for which B_i is empty then

$$S(p,k) = |\bar{A}_1 \cap \bar{A}_2 \cap \cdots \cap \bar{A}_k|, \quad \text{and} \quad |U| = k^p,$$

since each of the p elements can be put into any one of the k distinguishable boxes. Let t be an integer such that $1 \leq t \leq k$. How many partitions of U belong to the intersection $A_1 \cap A_2 \cap \cdots \cap A_t$? For these partitions the boxes $B_1, \ldots, B_t$ are empty and the remaining boxes $B_{t+1}, \ldots, B_k$ may or may not be empty. Thus, $|A_1 \cap A_2 \cap \cdots \cap A_t|$ counts the number of partitions of $[p]$ into $k - t$ distinguishable boxes and hence equals $(k-t)^p$. The same conclustion holds no matter which t boxes are assumed empty; that is

$$|A_{i_1} \cap A_{i_2} \cap \cdots \cap A_{i_t}| = (k-t)^p$$

for each t-subset $\{i_1, i_2, \ldots, i_t\}$ of $\{1, 2, lodots, k\}$. By the inclusion-exclusion principle, we have

$$S(p,k) = \sum_{t=0}^{k} (-1)^t \binom{k}{t} (k-t)^p,$$

and we are done. $\square$